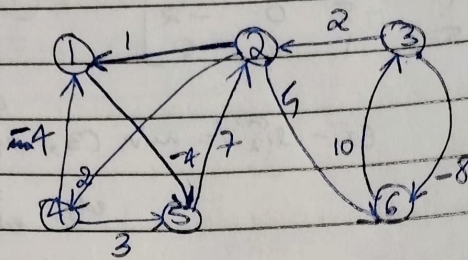


Q. Execute faster all pair shortest path algo. on the following graph from every vertex to every other vertices.



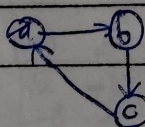
$$L^{(6)} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} 0 & 3 & \infty & \infty & -4 & \infty \\ -2 & 0 & 15 & 2 & -3 & 5 \\ 3 & 2 & 0 & 4 & \infty & -8 \\ -4 & 10 & \infty & 0 & -8 & \infty \\ 8 & 7 & \infty & 9 & 0 & 12 \\ \infty & 12 & 10 & \infty & \infty & 0 \end{bmatrix} \end{matrix}$$

$$L^{(6)} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} 0 & & & & & \\ & 0 & & & & \\ & & 0 & & & \\ & & & 0 & & \\ & & & & 0 & \\ & & & & & 0 \end{bmatrix} \end{matrix}$$

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⊕ Transitive closure.

" " of a directed graph G is defined as $n \times n$ boolean matrix $T = t_{ij}$, in which the element is i^{th} row and j^{th} col, i.e. $t_{ij} = 1$ if there exists a non-trivial directed path. (path w/o cycle) directed path of the length 1 otherwise it is zero.



$$T = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

find a path from every vertex to every other vertex.

(not shortest path)

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(Floyd algorithm)

Transitive closure (G):

→ he weights given

$n \leftarrow$ rows of G .

for $i=1$ to n

for $j=1$ to n

if $(i=j)$ or $i, j \in E$

else

$t_{ij} = 1$

$t_{ij} = 0$.

for $k=1$ to n

for $i=1$ to n

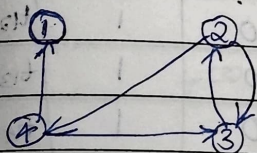
for $j=1$ to n

$t_{ij}^{(k)} = (t_{ij}^{(k-1)}) \vee (t_{ik}^{(k-1)} \wedge t_{kj}^{(k-1)})$

return $T(n)$

$O(n^3)$

k is an intermediate vertex in set 1 to n .



$V = \{1, 2, 3, 4\}$.

	1	2	3	4
1	1	0	0	0
2	0	1	1	1
3	0	1	1	0
4	1	0	1	1

[only mark 1 when $i=j$ or there is a direct path]

k	i	j	$t_{ij}^{(k-1)}$	$t_{ik}^{(k-1)}$	$t_{kj}^{(k-1)}$	Result: $t_{ij}^{(k)}$
1	1	1	1	$t_{11}^{(0)} = 1$	$t_{11}^{(0)} = 1$	$t_{11} = t_{11} \vee (t_{11} \wedge t_{11}) = 1 \vee (1 \wedge 1) = 1$
		2	0	$t_{12}^{(0)} = 1$	$t_{12}^{(0)} = 0$	$t_{12} = 0 \vee (t_{11} \wedge t_{12}) = 0 \vee (1 \wedge 0) = 0$
		3	0	$t_{13}^{(0)} = 1$	$t_{13}^{(0)} = 0$	$t_{13} = 0 \vee (1 \wedge 0) = 0$
		4	0	$t_{14}^{(0)} = 1$	$t_{14}^{(0)} = 0$	$t_{14} = 0 \vee (1 \wedge 0) = 0$
1	2	1	0	$t_{21}^{(0)} = 0$	$t_{21}^{(0)} = 1$	$t_{21} = 0 \vee (0 \wedge 1) = 0$
		2	1	0	$t_{22}^{(0)} = 0$	$t_{22} = 1 \vee (0 \wedge 0) = 1$
		3	1	0	$t_{23}^{(0)} = 0$	$t_{23} = 1 \vee (0 \wedge 0) = 1$
		4	1	0	$t_{24}^{(0)} = 0$	$t_{24} = 1 \vee (0 \wedge 0) = 1$

k	i	j	$t_{ij}^{(k-1)}$	$t_{ik}^{(k-1)}$	$t_{kj}^{(k-1)}$	$t_{ij}^{(k)}$
1	3	1	0	$t_{31}^{(0)} = 0$	1	$t_{31}^{(1)} = 0 \vee (0 \wedge 0) = 0$
		2	1	0	0	$t_{32}^{(1)} = 1 \vee (0 \wedge 0) = 1$
		3	1	0	0	$t_{33}^{(1)} = 1 \vee (0 \wedge 0) = 1$
		4	0	0	0	$t_{34}^{(1)} = 0 \vee (0 \wedge 0) = 0$
1	4	1	1	$t_{41}^{(0)} = 1$	1	$t_{41}^{(1)} = 1 \vee (0 \wedge 0) = 1$
		2	0	1	0	$t_{42}^{(1)} = 0 \vee (1 \wedge 0) = 0$
		3	1	1	0	$t_{43}^{(1)} = 1 \vee (1 \wedge 0) = 1$
		4	1	1	0	$t_{44}^{(1)} = 1 \vee (1 \wedge 0) = 1$

$$T^{(0)} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

k	i	j	$t_{ij}^{(k-1)}$	$t_{ik}^{(k-1)}$	$t_{kj}^{(k-1)}$	$t_{ij}^{(k)}$
2	1	1	1	$t_{12}^{(0)} = 0$	0	$t_{11} = 1 \vee (0 \wedge 0) = 1$
		2	0	0	1	$t_{12} = 0 \vee (0 \wedge 1) = 0$
		3	0	0	1	$t_{13} = 0 \vee (0 \wedge 1) = 0$
		4	0	0	1	$t_{14} = 0 \vee (0 \wedge 1) = 0$
2	2	1	0	$t_{21}^{(0)} = 1$	0	$t_{21} = 0 \vee (1 \wedge 0) = 0$
		2	1	1	1	$t_{22} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{23} = 1 \vee (1 \wedge 1) = 1$
		4	1	1	1	$t_{24} = 1 \vee (1 \wedge 1) = 1$
2	3	1	0	$t_{32}^{(0)} = 1$	0	$t_{31} = 0 \vee (1 \wedge 0) = 0$
		2	1	1	1	$t_{32} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{33} = 1 \vee (1 \wedge 1) = 1$
		4	0	1	1	$t_{34} = 0 \vee (1 \wedge 1) = 1$
2	4	1	1	$t_{42}^{(0)} = 0$	0	$t_{41} = 1 \vee (0 \wedge 0) = 1$
		2	0	0	1	$t_{42} = 0 \vee (0 \wedge 1) = 0$
		3	1	0	1	$t_{43} = 1 \vee (0 \wedge 1) = 1$
		4	1	0	1	$t_{44} = 1 \vee (0 \wedge 1) = 1$

$$T^{(2)} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

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k	i	j	$t_{ij}^{(k-1)}$	$t_{ik}^{(k-1)}$	$t_{kj}^{(k-1)}$	$t_{ij}^{(k)}$
3	1	1	1	0	0	$t_{11} = 1 \vee (0 \wedge 0) = 1$
		2	0	0	0	$t_{12} = 0 \vee (0 \wedge 0) = 0$
		3	0	0	1	$t_{13} = 0 \vee (0 \wedge 0) = 0$
		4	0	0	1	$t_{14} = 0 \vee (0 \wedge 0) = 0$
3	2	1	0	1	0	$t_{21} = 0 \vee (1 \wedge 0) = 0$
		2	1	1	1	$t_{22} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{23} = 1 \vee (1 \wedge 1) = 1$
		4	1	1	1	$t_{24} = 1 \vee (1 \wedge 1) = 1$
3	3	1	0	1	0	$t_{31} = 0 \vee (1 \wedge 0) = 0$
		2	1	1	1	$t_{32} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{33} = 1 \vee (1 \wedge 1) = 1$
		4	1	1	1	$t_{34} = 1 \vee (1 \wedge 1) = 1$
3	4	1	1	1	0	$t_{41} = 1 \vee (1 \wedge 0) = 1$
		2	0	1	1	$t_{42} = 0 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{43} = 1 \vee (1 \wedge 1) = 1$
↑		4	1	1	1	$t_{44} = 1 \vee (1 \wedge 1) = 1$

$$T^{(3)} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

k	i	j	$t_{ij}^{(k-1)}$	$t_{ik}^{(k-1)}$	$t_{kj}^{(k-1)}$	$t_{ij}^{(k)}$
4	1	1	1	0	1	$t_{11} = 1 \vee (0 \wedge 1) = 1$
		2	0	0	1	$t_{12} = 0 \vee (0 \wedge 1) = 0$
		3	0	0	1	$t_{13} = 0 \vee (0 \wedge 1) = 0$
		4	0	0	1	$t_{14} = 0 \vee (0 \wedge 1) = 0$
4	2	1	0	1	1	$t_{21} = 0 \vee (1 \wedge 1) = 1$
		2	1	1	1	$t_{22} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{23} = 1 \vee (1 \wedge 1) = 1$
		4	1	1	1	$t_{24} = 1 \vee (1 \wedge 1) = 1$
4	3	1	0	1	1	$t_{31} = 0 \vee (1 \wedge 1) = 1$
		2	1	1	1	$t_{32} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{33} = 1 \vee (1 \wedge 1) = 1$
		4	1	1	1	$t_{34} = 1 \vee (1 \wedge 1) = 1$

k	i	j	$t_{ij}^{(k-1)}$	$t_{ik}^{(k-1)}$	$t_{kj}^{(k-1)}$	$t_{ij}^{(k)}$
4	4	1	1	1	1	$t_{41} = 1 \vee (1 \wedge 1) = 1$
		2	1	1	1	$t_{42} = 1 \vee (1 \wedge 1) = 1$
		3	1	1	1	$t_{43} = 1 \vee (1 \wedge 1) = 1$
		4	1	1	1	$t_{44} = 1 \vee (1 \wedge 1) = 1$

	1	2	3	4
1	1	0	0	0
2	1	1	1	1
3	1	1	1	1
4	1	1	1	1

⊕ Floyd-Warshall algorithm (G, E, W)

$n \leftarrow \text{rows}(G)$

$W^{(0)} \leftarrow W$

for $k=1$ to n

for $i=1$ to n

for $j=1$ to n

$$d_{ij}^{(k)} = \min(d_{ij}^{(k-1)}, \min(d_{ik}^{(k-1)} + d_{kj}^{(k-1)}))$$

return $W^{(n)}$

$$W^{(0)} = \text{dis}^{(0)} = \begin{bmatrix} d_{11}^{(0)} & \dots & d_{1n}^{(0)} \\ \vdots & & \vdots \\ d_{n1}^{(0)} & \dots & d_{nn}^{(0)} \end{bmatrix}$$

$\uparrow$
 W

• k is an intermediate vertex in a simple path

$P = \{v_1, v_2, \dots, v_k\}$ and is any vertex v other than $v_1 \neq v_k$.

• $d_{ij}^{(k)}$ is the shortest path weight from vertex i to j for which all intermediate vertices are in the set 1 to k .

• If $k=0$, $d_{ij}^{(0)} = 0$ if $i=j$.

$= W_{ij}$ if $(i, j) \in E$ otherwise

$= \infty$ otherwise.

